

AI-Powered Intelligent Sales Forecasting Platform using Predictive Analytics

Group_1

TEAM B

Divya Bharathi I

Project Lead + Backend Developer

Jagadeesh

AI/ML Engineer

Guna Shekar Naidu Khandapu

Database / DevOps Engineer

Gopika Rajendiran

Frontend Developer

Gilagalla Karthik

QA / Test Engineer

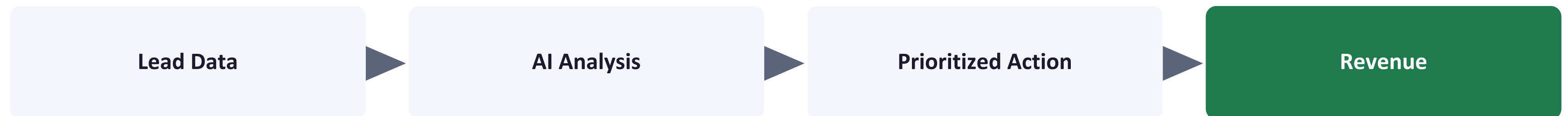
The problem, and what we built

The Problem

Sales teams spend significant time manually researching prospects, preparing outreach, and tracking interactions — leading to inconsistent engagement and lower conversion rates.

The Solution

SalesGenie AI automates lead analysis, prospect research, personalized outreach generation, sales conversation summarization, and follow-up recommendations — using LLMs, CRM integrations, and sales intelligence analytics — so reps focus on relationship building, not manual research.



What the platform delivers



Centralized Lead Database

Single source of truth for every prospect



AI Company Analysis

Automated qualification scoring & insights



Personalized Outreach

AI-generated emails tailored per lead



Automated Lead Scoring

Rules + AI-based prioritization engine



Conversation Intelligence

AI summaries + action items from calls



CRM Integration

Sync architecture ready for Salesforce/HubSpot



Sales Analytics Dashboard

Pipeline, conversion & KPI visibility



Automated Follow-Ups

Rule-based recommendation feed



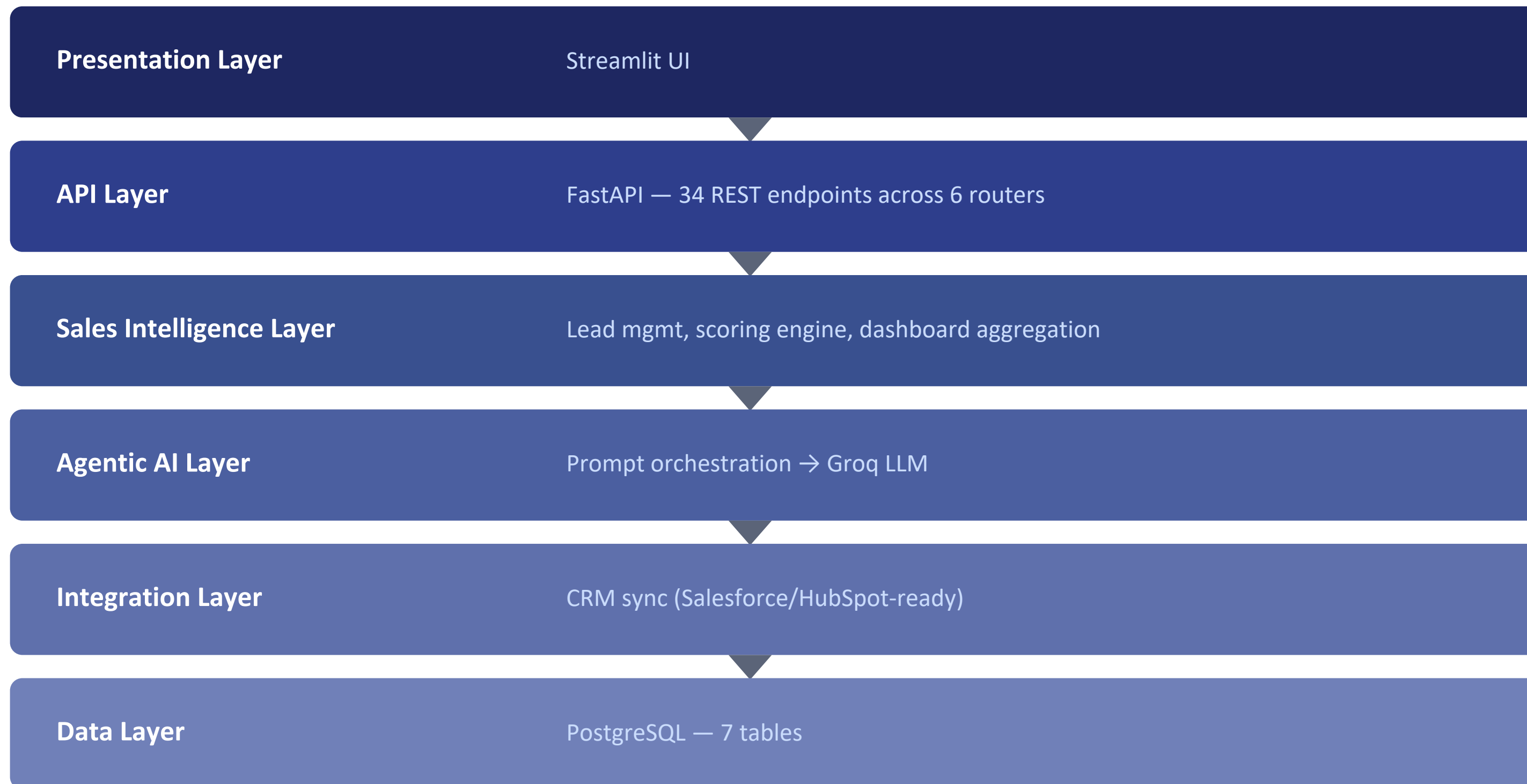
What we built it with, and how it flows

Backend	FastAPI (Python), Pydantic, SQLAlchemy ORM
Frontend	Streamlit
Database	PostgreSQL 16
AI / LLM Layer	Groq API, llama-3.3-70b-versatile

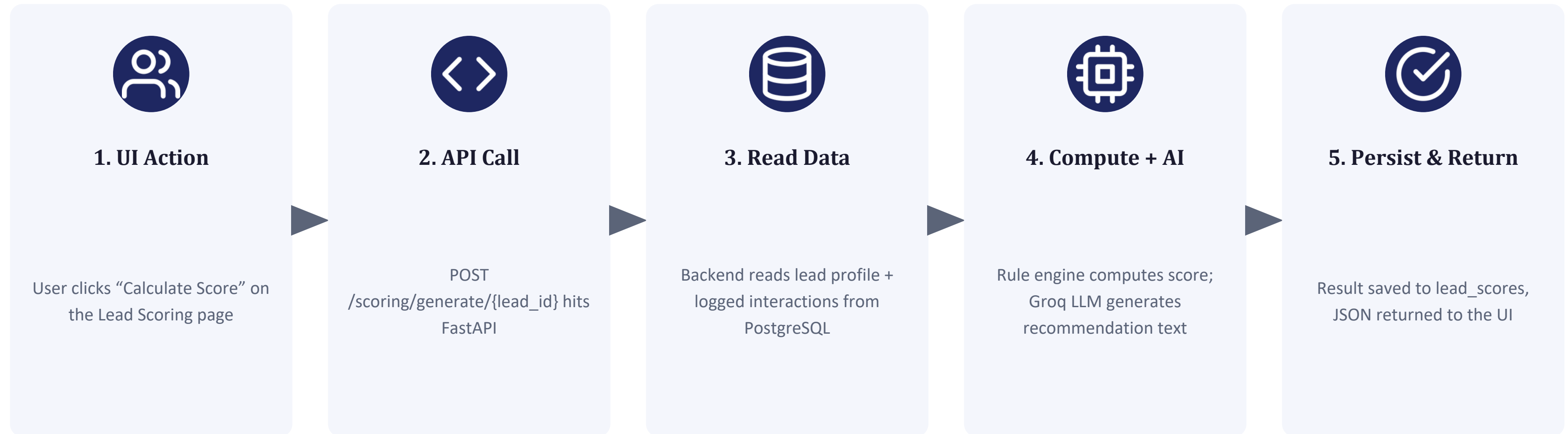
Development Workflow



Layered system architecture

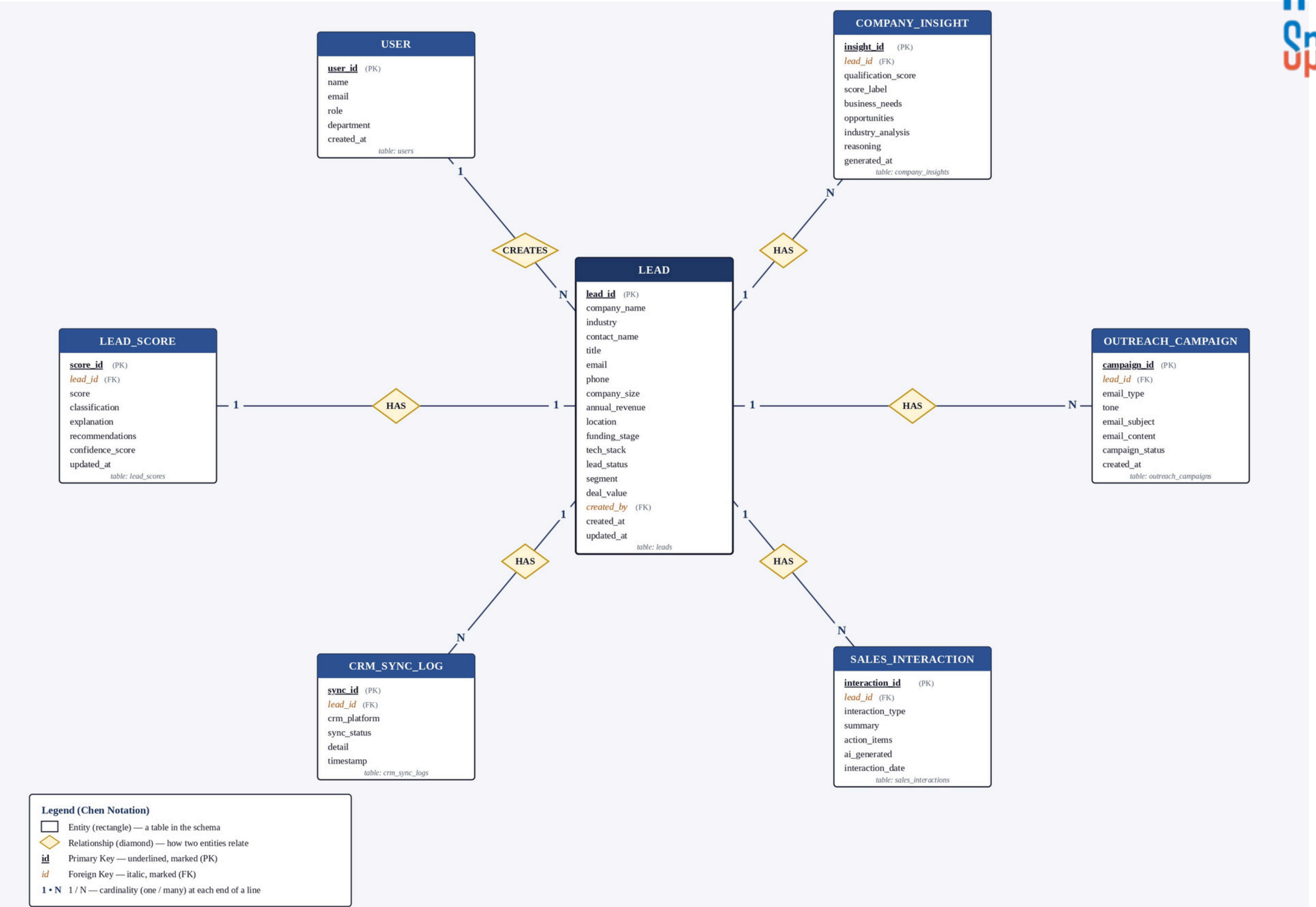


Request flow — example: Lead Scoring



Every module follows this same shape: UI → FastAPI endpoint → PostgreSQL (+ Groq LLM where AI is involved) → JSON response → UI renders it.

Database schema (ER diagram)





Module 1 — Lead Management & Prospect Database

Overview

Full CRUD for prospects, search/filter by company, contact, or status, engagement history logging, and bulk import via CSV or manual entry with automatic de-duplication (matches on email or company name).

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NAVIGATION

Dashboard

Lead Management

Add Lead

Lead Intelligence

AI Outreach

Lead Scoring

Conversations

Deploy

Lead workspace

Search, review, update, and track every prospect in one focused workspace.

Search leads

Company, contact, industry, or location

Matching leads

41

Active pipeline

41

Estimated value

\$5,876,000

Prospects

Company	Contact	Industry	Stage	Value
VisionX Media	Noah Richardson	Digital Media	Contacted	\$58,000
Harbor Freight Tech	Matthew Evans	Supply Chain	New	\$390,000
Raj tech	Bharathi	IT	Contacted	\$10,000
EcoHarvest Foods	Daniel Foster	Food Technology	Contacted	\$205,000
Forge Industrial	Sarah Mitchell	Industrial Automation	Negotiation	\$610,000
PureWater Solutions	Hannah Reed	Environmental Tech	Qualified	\$98,000
Aether Aerospace	Nathan Hughes	Aerospace	Contacted	\$520,000
Urban Mobility Labs	Arjun Mehta	Transportation	New	\$48,000

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Add Lead

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Add prospects

Create a detailed lead or import a prospect list in a few steps.

Add one lead

Import leads

Contact details

Company name *

Contact name

Job title

Email

Phone

Location

Business profile

Industry

Company size

Annual revenue

Funding stage

Estimated deal value

Segment

Add prospects

Create a detailed lead or import a prospect list in a few steps.

Add one lead

Import leads

Upload a CSV with a required company_name column, or paste a simple comma-separated list.

CSV upload

Paste rows

Choose CSV file

Upload

200MB per file • CSV

Import CSV



Module 4 — Lead Scoring & Recommendation Engine

Overview

Computes a score from firmographics AND engagement history (website visits, email opens/replies, demo bookings), classifies into 5 tiers (Platinum → Low Priority), and includes a What-If simulator plus AI recommendations.

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NAVIGATION

Dashboard

Lead Management

Add Lead

Lead Intelligence

AI Outreach

Lead Scoring

Conversations

Lead Scoring & Recommendations

Rank prospects and run simulated What-If analyses.

Priority Rankings

Score & AI Recommendations

What-If Simulator

Lead Priority Pipeline

Company	Score	Classification	Industry	Status
CloudScale Inc.	75	Silver	Cloud Infrastructure	Proposal
SecureNet Global	68	Silver	Network Security	Qualified
Apex Biotech	66	Silver	Biotechnology	New
Aether Aerospace	66	Silver	Aerospace	Contacted
Harbor Freight Tech	66	Silver	Supply Chain	New
FinEdge Solutions	60	Bronze	FinTech	Qualified
CloudVista Technologies	60	Bronze	Information Technology	New
FinEdge Capital	59	Bronze	FinTech	Proposal Sent
EcoHarvest Foods	59	Bronze	Food Technology	Contacted
ABC Technologies	58	Bronze	Software	New

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Conversations

Lead Scoring & Recommendations

Rank prospects and run simulated What-If analyses.

Priority Rankings

Score & AI Recommendations

What-If Simulator

Select Lead to Score

VisionX Media — Noah Richardson · SMB

Recalculate Score

Bronze

Lead score

58/100

Data confidence: 85%

VisionX Media

Digital Media · SMB

Company Size: 50-100

Annual Revenue: 7M - 10M

Location: Los Angeles

Funding Stage: Bootstrapped

Pipeline Stage: Contacted

Est. Deal Value: \$58,000

Recommended next steps

Recommendations for VisionX Media

Based on the prospect profile and detailed score breakdown, here are specific, actionable recommendations for the sales team:

Draft a personalized email to Ananya, targeting the pain points discussed in the meeting, such as unified pipeline view and data visibility, and highlighting how the proposed solution can help VisionX Media make data-driven decisions to optimize content creation and distribution processes. The tone should be consultative, emphasizing the value proposition of improved business

Score breakdown

Email Reply (Max 10)

Points

0.0 pts

Recent News (Max 10)

Points

10.0 pts

Company Size (Max 15)

Points

6.0 pts

Hiring Trend (Max 10)

Points

3.0 pts

Funding Stage (Max 10)

Points

1.0 pts

Annual Revenue (Max 15)

Points

12.0 pts

58



Module 5 — Conversation Intelligence & CRM Integration

Overview

Syncs a lead's profile to Salesforce or HubSpot (simulated — no live OAuth credentials configured, but the audit-trail contract matches a real integration), and summarizes call/meeting transcripts into key points and action items via AI.

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NAVIGATION

Dashboard

Lead Management

Add Lead

Lead Intelligence

AI Outreach

Lead Scoring

Conversations

Deploy

Conversations

CRM sync status and AI-powered meeting/call summarization.

Select Lead

VisionX Media — Noah Richardson · SMB

CRM Sync Status

Meeting Summary (AI Powered)

Recent Activity

VisionX Media — sync this lead's current profile to a CRM platform.

Simulated integration: no live Salesforce/HubSpot credentials are configured for this project, so this logs the same sync record a real integration would produce.

CRM Platform

Salesforce

Sync Now

Sync History

Platform	Status	Detail	Timestamp
HubSpot	Synced	Contact 'Noah Richardson' and deal stage 'Contacted' pushed to HubSpot.	2026-08-17T06:00:40.701595
Salesforce	Synced	Contact 'Noah Richardson' and deal stage 'Contacted' pushed to Salesforce.	2026-08-17T06:00:34.007981
Salesforce	Synced	Contact 'Noah Richardson' and deal stage 'Contacted' pushed to Salesforce.	2026-07-30T20:46:34.109874

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CRM Sync Status

Meeting Summary (AI Powered)

Recent Activity

Type

Call

Meeting

Paste the call/meeting transcript

Rep: Hi Sarah, thanks for joining...

Sarah: We're looking for better data pipeline monitoring...

Generate Summary

Past AI Summaries for this Lead

Meeting — 2026-07-30T20:46:10.602165

The sales rep discussed NovaRetail's need for a unified pipeline view, automated lead prioritization, and better follow-up tracking with Ananya. Ananya expressed interest in AI-generated outreach, but prioritized data visibility and manager reporting. The prospect is evaluating solutions and aims to shortlist vendors this month. A proposal and product demo are expected next steps.

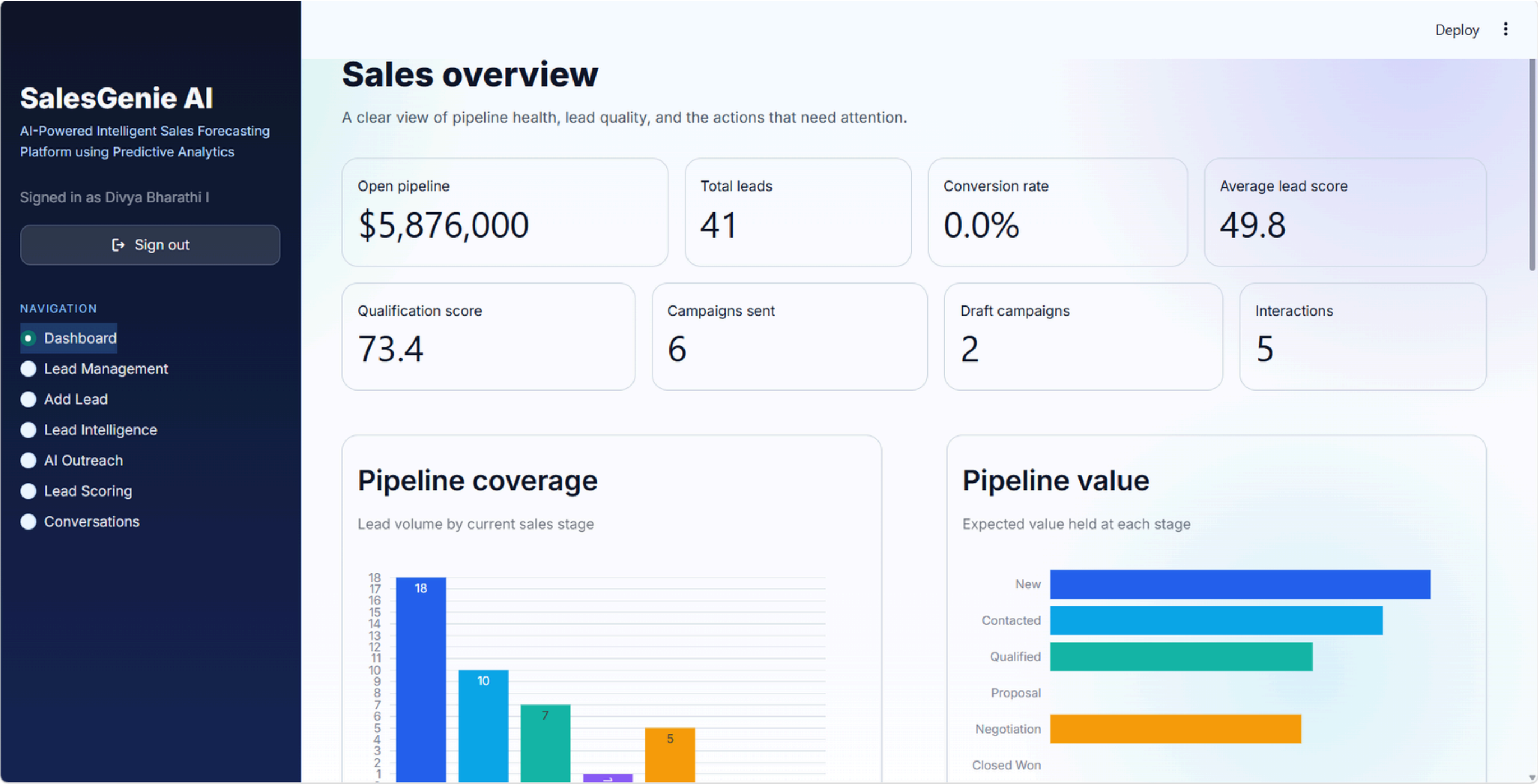
Send proposal with implementation timeline, integration requirements, and pricing | Arrange product demo for next Tuesday with Head of Sales Operations included | Follow up with Ananya and Head of Sales Operations after demo



Module 6 — Dashboard & Sales Analytics

Overview

Aggregates KPIs (conversion rate, pipeline value, avg scores), a Kanban-style pipeline board by stage, and a rule-based automated follow-up feed — no LLM call needed, so it's always instant.



Testing results

34

Automated tests

pytest, all 6 modules

31 / 34

Passed

0 failures, 0 errors

100%

Scoring accuracy

within-one-tier, vs ≥85% target

Methodology

- 3 skipped tests are AI-generation endpoints — skipped with a stated reason (no LLM network access in the test environment), not falsely reported as failures.
- Lead scoring validated with 7 synthetic leads across 4 industries, each with matching engagement history — not firmographics alone.
- Accuracy measured against reasoned QA judgment — documented explicitly as a known limitation, not overstated.

Challenges & Learnings



Challenges

- Integrating multiple modules into a single platform while maintaining smooth communication between the frontend, backend, AI services, database, and CRM systems.
- Designing a scalable backend architecture to support lead management, AI-powered outreach, lead scoring, CRM synchronization, and conversation intelligence.
- Mapping different CRM schemas (Salesforce and HubSpot) to a unified internal data model while preserving data consistency.
- Coordinating development, integration, testing, and presentation activities across a five-member team within project deadlines.

Key Learnings

- Gained hands-on experience in building REST APIs using FastAPI and integrating them with PostgreSQL through SQLAlchemy.
- Learned how to design and implement AI-assisted features
- Improved debugging, API testing, database design, version control, and deployment skills.

Thank you